

Policies have prose reviews, but no unit tests

A prototype for turning session-confirmed travel & expense intent into traceable scenarios and deterministic checks.

SYNTHETIC PROTOTYPE · VALIDATION PENDING

Narrow claim

**Make ambiguity
executable — then test it.**

One owner carries every edge case in their head

FINANCE / PEOPLE OPS

Policy owner

Maintains rules, fields exceptions, and explains decisions across many combinations of amount, category, timing, destination, and prior spend.

HYPOTHESIS — NOT USER RESEARCH

1

Thresholds hide boundary gaps

Manual review can miss equality cases and adjacent ranges.

2

Exceptions collide across clauses

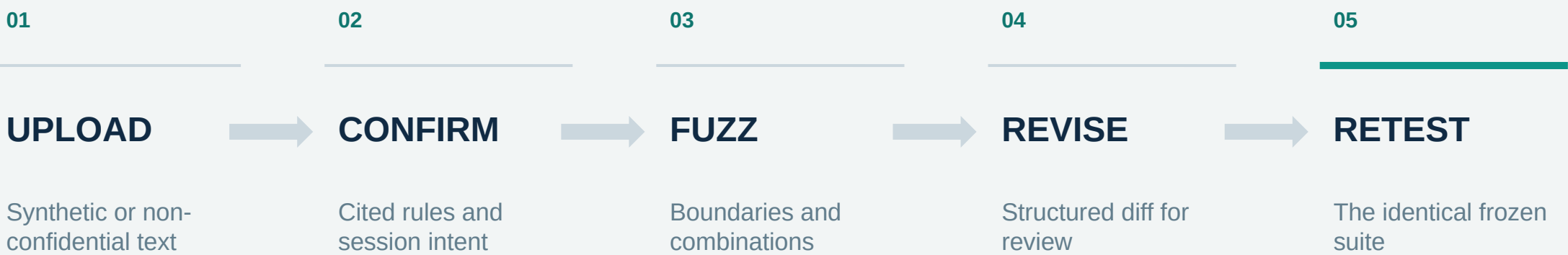
Priority and overrides may exist only in reviewers' memory.

3

Change review lacks a fixed regression set

A wording fix can shift a different outcome without a visible trace.

The journey ends in a retest, not a summary

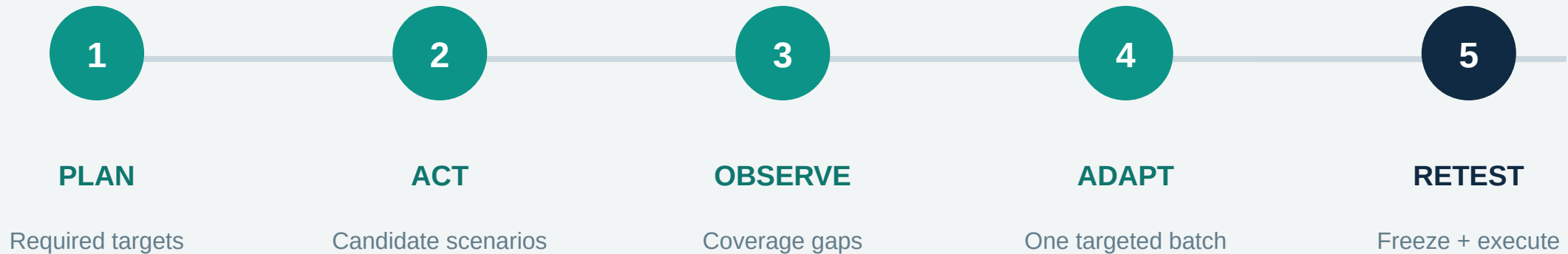


Why it differs from summarization

Every finding should resolve to visible facts, a deterministic trace, and an exact source citation — then survive the same-suite comparison.

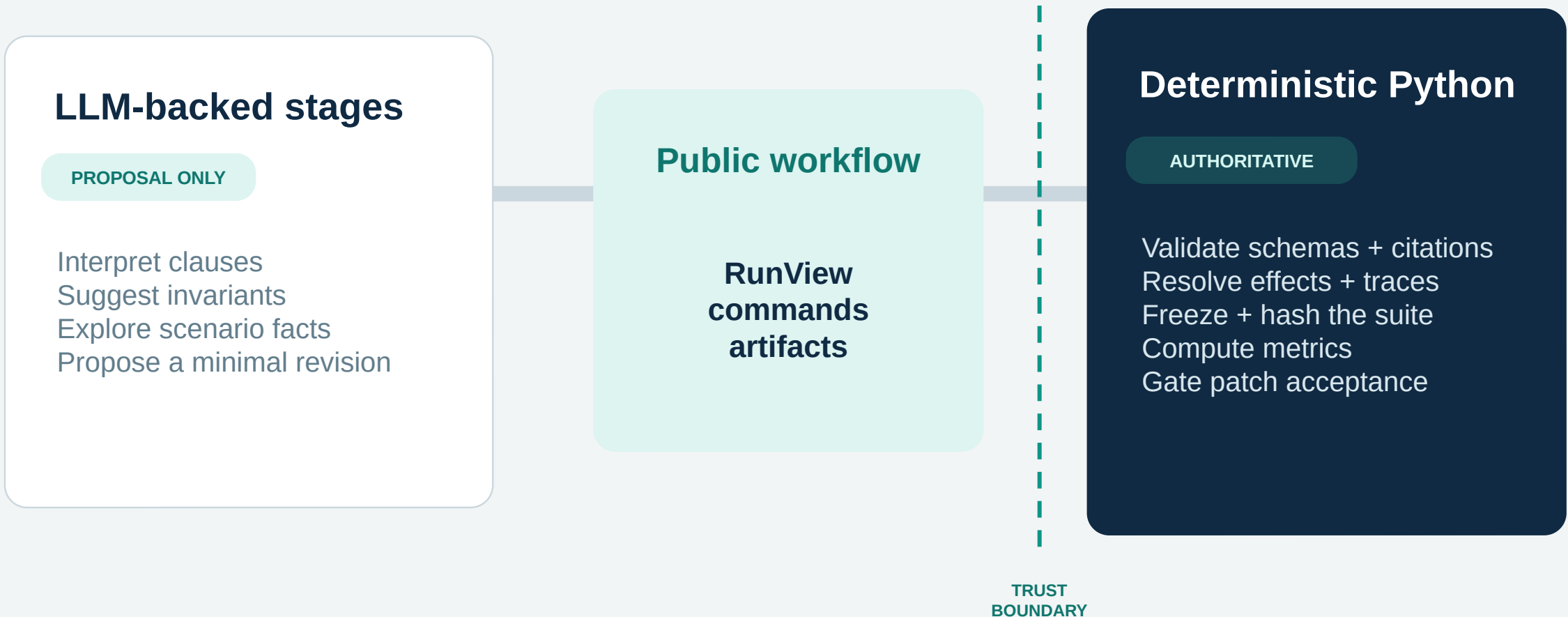
LIVE / CACHED MODE STAYS VISIBLE

One targeted cycle closes coverage — or stops

**Bound**

At most one targeted generation cycle before suite freeze.

Models propose; deterministic Python decides



Three planned cases make ambiguity executable

01	THRESHOLD GAP SGD 50.00 receipt claim	below 50 ≠ above 50 Equality has no matching rule	TRACE + CITATION PENDING
02	HOTEL CONFLICT International hotel · SGD 249.00	no approval ↔ manager approval No override resolves the clash	TRACE + CITATION PENDING
03	DAILY CAP BREACH Meal SGD 60 + prior SGD 50	eligible per claim → daily total 110 Confirmed daily intent is breached	TRACE + CITATION PENDING

PLANNED Final witnesses, authoritative traces, and exact policy citations will be captured only from the verified build.

Measured results wait for a frozen evidence chain

MEASURED RESULTS: PENDING

DEVELOPMENT

Known synthetic defects

- Exact finding match
- Citation integrity
- Coverage + replay
- Same-suite patch gates

CORRECTED CONTROL

Regression challenge

- No confirmed target defect
- Protected assertions
- No new gaps or conflicts
- Same engine + suite

BLIND

Late-sealed candidate

- Human review pending
- Scoring-ineligible
- First-run custody required
- No headline claim

Release chain **Gate B report** → **demo-core-v1** → **6-hour freeze** → **evidence capture** → **reviewer checks**

Results will be reported with their frozen suite, engine, and source evidence.

The impact hypothesis is better review evidence

TODAY

Policy review relies on memory, spot checks, and examples that change between reviewers.

NO ROI OR SAVINGS CLAIM BEFORE MEASUREMENT

Future proof
points

time to first actionable finding · total run time · confirmations · model calls · provider cost or N/A

HYPOTHESIS

1

Broader review coverage

Systematic boundaries and combinations expose what was exercised.

2

Traceable decisions

Facts, rule paths, and citations make disagreements inspectable.

3

Regression evidence

The same frozen suite tests a confirmed structured change.

A narrow prototype with a clear finish line

The intended proof

A session-confirmed structured change fixes targeted failures and introduces no protected regressions in the disclosed frozen suite.

VALIDATION PENDING

Next: capture verified results and footage after Gate B.

Limits

Prototype; validation pending
Synthetic T&E scope
No legal-compliance claim
No automatic publication
Human confirmation remains required

Five-person ownership

P1 integration · P2 policy · P3 fuzzing
P4 evaluation · P5 product

github.com/Luigi-Simon/policyfuzz